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~~Sınıf~~

for blue:

③ $\Delta X = \frac{n\lambda L}{d}$

$\Delta x_{blue} = \frac{1 \times 5 \times 470 \times 10^{-9}}{100 \times 10^{-6}} = 0.0235\ m$

d ← distance between sources

$n = 1$

for red

$\Delta x_{red} = \frac{1 \times 5 \times 680 \times 10^{-9}}{100 \times 10^{-6}} = 0.034\ m$

Seperation → $\Delta x_{red} - \Delta x_{blue} = 0.034 - 0.0235 = 0.0105\ m$

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① angle of incidence → 30°

$n_1 = 1$

$n_2 = 1.8$

$R_{TM} = \left[\frac{n_1 \cos\theta_1 - n_2 \cos\theta_2}{n_1 \cos\theta_1 + n_2 \cos\theta_2}\right]^2$

$\theta_1 = 30$

$\theta_2 = 64.16$

$R_{TM} = 0.0738$

$T = 1 - R$

$1 - 0.0738$

$T = 0.9262$

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⑤ concave surface $R = 45cm$

image → 35cm from real face

$f = -\frac{R}{2} = -22.5$

$d_i = 35\ cm$



$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

$\frac{1}{-22.5} = \frac{1}{d_o} + \frac{1}{35}$

$d_o = -13.69\ cm$

Scanned with CamScanner

(4)

angle between the transmission axes and the vertical → 20°

projection of polarized light in transmission axes → $\cos(20) = 0.94$

intensity of light in first polarizer $\frac{800}{2} \times 0.94 = 376\ W/m^2$

at 50°:

$\cos 50 = 0.64$

$\frac{376}{2} \times 0.64 = 120.3\ W/m^2$

at 140°

$\cos 140 = -0.77$ → Negative sign → opposite direction

$\frac{120.3}{2} \times 0 = 0\ W/m^2$

Scanned with CamScanner